

Practice Exam 1 • Answer Key

Chemistry Interim • 80 points • Unit 1: 48 points, Unit 2: 16 points, Unit 3: 16 points

Part A • Multiple Choice (2 points each, 36 total: Unit 1 questions 1–11, Unit 2 questions 12–15, Unit 3 questions 16–18)

Quick key: 1 B 2 D 3 C 4 A 5 D 6 B 7 C 8 C 9 B 10 A 11 D 12 A 13 B 14 A 15 D 16 C 17 B 18 A

#	Ans	Why
1	B	Thomson. The cathode ray bent toward a positive plate, so it was made of negative particles: electrons. Rutherford found the nucleus; Chadwick found the neutron.
2	D	most of the atom is empty space. Particles that hit nothing go straight through, so most of the atom is empty space. The few that bounced back showed the nucleus.
3	C	works only for atoms with one electron, such as hydrogen. Bohr's model predicts the hydrogen spectrum but fails for atoms with more than one electron. A describes Thomson's model; D is Dalton's.
4	A	neutron. The neutron is neutral with a mass close to the proton's. The electron's mass is about 1/1836 amu.
5	D	20. Neutrons = mass number – atomic number = 39 – 19 = 20. 19 is the number of protons.
6	B	$^{32}_{16}\text{S}^{2-}$. 16 protons means sulfur. Mass number = 16 + 16 = 32. Charge = 16 – 18 = 2–.
7	C	the same number of protons but different numbers of neutrons. Same protons (same element), different neutrons (different mass number). D describes ions.
8	C	10.8 amu. $10 \times 0.20 + 11 \times 0.80 = 10.8$ amu. The average sits closest to the more abundant isotope.
9	B	two light nuclei combine to form a heavier nucleus. Fusion joins light nuclei (as in the Sun). A is fission; C is ionic bonding; D is light emission.
10	A	$1s^2 2s^2 2p^6 3s^2 3p^3$. 15 electrons in filling order. B skips 3p; C overfills 2p (it holds 6); D is an excited state.
11	D	falls from a higher energy level to a lower one. Energy is absorbed on the way up and released as a photon on the way down.
12	A	Na. All are in period 3. Radius decreases across a period, so the element furthest left (Na) is largest.
13	B	decreases, because the outer electron is farther from the nucleus and more shielded. Each row adds an energy level. The outer electron is farther away and shielded by more inner electrons, so it is easier to remove.
14	A	fluorine. Electronegativity increases across a period and up a group, so fluorine (top right) is highest.
15	D	larger, because the extra electron adds repulsion while the nuclear charge stays the same. An anion is larger than its atom: same 17 protons, one more electron, more repulsion.

16	C	MgO. A metal (Mg) with a non-metal (O); $\Delta EN = 3.5 - 1.2 = 2.3$. The other three contain only non-metals.
17	B	iron(III) chloride. Three Cl^- make $3-$, so iron is Fe^{3+} . Iron has more than one charge, so the Roman numeral is needed. No prefixes for ionic compounds.
18	A	CO₂. CO ₂ is linear with two identical C=O dipoles that cancel. H ₂ O and NH ₃ have lone pairs on the central atom; HCl has one polar bond.

Part B • Free Response (44 points)

19. Atomic models (8 points)

(a) 1 point per row ($\frac{1}{2}$ for each cell).

Scientist	Experiment or evidence	What was discovered or proposed
Thomson	cathode ray tube	the electron ; "plum pudding" model (electrons in a positive sphere)
Rutherford	gold foil (alpha scattering) experiment	the nucleus ; tiny, dense, positive; the atom is mostly empty space
Bohr	line spectrum of hydrogen	electrons in fixed energy levels ; light is emitted when an electron drops between levels
Chadwick	beryllium bombarded with alpha particles	the neutron

(b) Any two, each with its conclusion: most alpha particles passed **straight through**, so the atom is **mostly empty space** · a few were **deflected**, so the positive charge is concentrated in one small region · very few **bounced back**, so the nucleus is very small, dense and positive.
1 point for each observation paired with its correct conclusion.

(c) Problem: an electron orbiting the nucleus should **lose energy** and spiral into the nucleus, so the atom would collapse (also accept: it cannot explain line spectra). Bohr's fix: electrons occupy **fixed energy levels** and do not lose energy while they stay in a level.
1 point: the problem. 1 point: how fixed energy levels solve it.

20. Atoms, isotopes and average atomic mass (10 points)

(a) 1 point per fully correct row.

Species	Protons	Neutrons	Electrons	How
${}_{13}^{27}\text{Al}^{3+}$	13	14	10	$n = 27 - 13$; $3+$ means 3 electrons lost
${}_{17}^{35}\text{Cl}^-$	17	18	18	$A = 17 + 18 = 35$; charge = $17 - 18 = 1-$
${}_{7}^{15}\text{N}$	7	8	7	neutral, so electrons = protons
potassium-41	19	22	19	$Z = 19$; $n = 41 - 19$

(b) $(23.985 \times 0.7899) + (24.986 \times 0.1000) + (25.983 \times 0.1101) = 18.946 + 2.499 + 2.861 = 24.31 \text{ amu}$

1 point: percents changed to fractions. 1 point: three products added. 1 point: answer with unit (accept 24.30 to 24.31).

(c) Let x = fraction of Cl-35, so $(1 - x)$ = fraction of Cl-37.

$$34.969x + 36.966(1 - x) = 35.453$$

$$36.966 - 1.997x = 35.453, \text{ so } x = 1.513 \div 1.997 = 0.7576$$

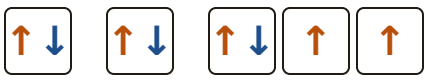
Cl-35: 75.8% · Cl-37: 24.2%

1 point: set-up with x and $(1 - x)$. 1 point: correct algebra. 1 point: both percents. Check: the average is nearer 35, so Cl-35 must be the larger share.

21. Electrons and light (8 points)

(a) Full: **$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$** · Noble gas: **$[\text{Ar}] 4s^2$**

1 point each. The superscripts must add to 20.

(b)  **2 unpaired electrons**

1 point: 8 arrows with 2p filled as one pair and two single arrows of the same spin (Hund's rule), opposite spins in every pair. 1 point: 2 unpaired.

(c) $2 + 2 + 6 + 1 + 1 = 12$ electrons, so the element is **magnesium**. It is in an **excited state**: the ground state is $1s^2 2s^2 2p^6 3s^2$, but here one electron is in 3p while 3s is not full.

1 point: magnesium. 1 point: excited, with the reason.

(d) Hydrogen and sodium. All four hydrogen lines (410, 434, 486, 656 nm) and all four sodium lines (498, 569, 589, 616 nm) appear in the unknown. Helium is absent: its lines (for example 447, 471 and 668 nm) are missing. Lithium is absent: only one of its lines (497 nm) is near a line in the unknown, and 460, 610 and 671 nm are missing.

1 point: both elements identified and no others. 1 point: reasoning that every line of an element must match.

22. Periodic trends (8 points)

(a) **$\text{Cl} < \text{S} < \text{Mg} < \text{Na}$**

2 points for the full order; 1 point if exactly one pair is swapped. All four are in period 3, so the radius decreases left to right.

(b) Across the period the number of **protons increases**, while the electrons are added to the **same energy level**, so the **shielding stays about the same**. The stronger nuclear pull draws the outer electrons closer.

1 point: more protons / greater nuclear charge. 1 point: same energy level or same shielding, so electrons are pulled closer. No credit for "more electrons make it bigger".

(c) Li. Its outer electron is in the 2nd energy level, **closer to the nucleus** and **less shielded** than potassium's outer electron in the 4th level, so it is held more tightly and takes more energy to remove.

1 point: Li. 1 point: distance and shielding.

(d) The Na atom is larger. Na^+ has **lost its outer (3s) electron**, so it has one fewer energy level, and the same 11 protons now pull on only 10 electrons.

1 point: atom larger. 1 point: lost the outer level / same protons, fewer electrons.

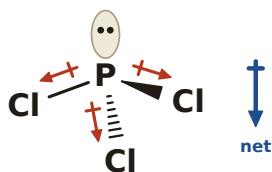
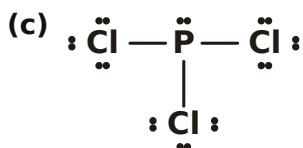
23. Chemical bonding (10 points)

(a) 1 point per row.

Bond	ΔEN	Type of bond
Na-F	$4.0 - 0.9 = 3.1$	ionic
O-H	$3.5 - 2.1 = 1.4$	polar covalent
Cl-Cl	$3.0 - 3.0 = 0$	nonpolar covalent

(b) CaBr_2 **calcium bromide** · CuO **copper(II) oxide** · N_2O_5 **dinitrogen pentoxide**

1 point each. No prefixes in an ionic name; copper needs its Roman numeral (O is 2−, so Cu is 2+).



26 valence electrons. 4 electron domains on P (3 bonds + 1 lone pair): electron geometry **tetrahedral**, molecular geometry **trigonal pyramidal**. The molecule is **polar**: the P-Cl bonds are polar (ΔEN 0.9) and the lone pair makes the shape unsymmetric, so the dipoles do not cancel.

1 point: Lewis structure with the lone pair on P and 3 lone pairs on each Cl. 1 point: tetrahedral. 1 point: trigonal pyramidal. 1 point: polar.